

# Epistemological and social beliefs are associated with support for different open science practices across the social sciences

Hannah Hok

[hok1@mit.edu](mailto:hok1@mit.edu)

MIT <https://orcid.org/0000-0003-2902-4201>

Micah Altman

MIT

Sakaria Auelua-Toomey

University of Hawaiʻi at Mānoa

Chris Bourg

MIT

---

## Article

### Keywords:

**Posted Date:** September 25th, 2026

**DOI:** <https://doi.org/10.21203/rs.3.rs-11134352/v1>

**License:**   This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

**Additional Declarations:** There is **NO** Competing Interest.

---

# Epistemological and social beliefs are associated with support for different open science practices across the social sciences

## Abstract

Open science comprises a range of practices that advance different goals, yet it is often treated as an expression of a single orientation toward openness. We ran a pre-specified, stratified survey of researchers at 292 U.S. universities to examine whether researchers' beliefs about scientific knowledge and scientific communities were associated with attitudes toward data sharing, preregistration, open access, and replication across eight social science disciplines. More positivist beliefs were associated with stronger support for data sharing, preregistration, and replication, and with weaker support for open access. Perceiving science as exclusionary was associated with stronger support for open access and data sharing. These relationships held after adjustment for disciplinary affiliation and methodological orientation. The findings indicate that open science is not supported for a single reason: different practices appeal to distinct but overlapping epistemic and social commitments.

Open science (OS) practices such as data sharing, preregistration, and open access have become increasingly visible and valued across academia. Yet despite growing consensus on their importance, OS is not a single practice or goal: its component practices advance multiple goals, and support for both practices and goals varies across fields. A wide range of data documents substantial differences in open science behaviors across and within disciplines. For instance, the National Academies of Sciences (NASEM 2018) report that the proportion of articles published as open access ranges from under 20% in chemistry to nearly 60% in the biomedical sciences. Piwowar et al. (2018) find even greater within-discipline variation: in biomedical sciences, subfields like fertility and tropical medicine adopt open access at rates over 80%, far above their parent discipline's average.

These differences are not confined to publishing. Attitudes and behaviors around data sharing and preregistration also vary considerably. However, while the patterns are increasingly well-documented, less is known about the association of researchers' beliefs about science with support for specific forms of openness. Some meta-science research points to field-level differences in infrastructure, incentives, or policies (NASEM 2018); others emphasize disciplinary histories. For example, political science and economics developed formal data infrastructures earlier than many other fields (Desrosières 2002; Bisco 1966) and were among the first to normalize article replication practices (King 1995; Anderson et al. 2008). In contrast, open-access preprint norms originated in physics with the founding of arXiv and spread later to other disciplines (Smart 2022). In short, disciplines are structured very differently--by culture, method, and infrastructure -- which makes isolating any single causal factor challenging. These

differences may shape not only whether researchers can implement particular practices, but also which goals of open science they value and which practices they support.

One particularly promising source of insight comes from a recent and extensive study of attitudes and behaviors of open science practice within the social sciences by Ferguson et al. (2023). Their survey captures a broad sample of PhD students and published authors across four social science disciplines (e.g., Economics, Political Science, Psychology, and Sociology) and measures multiple dimensions of OS behavior and belief, including awareness, attitudes, norms, and lifetime practice. Their findings reveal significant disciplinary differences in OS engagement, even within a single broad 'school' (social sciences). Data sharing was reported at roughly twice the rate of preregistration across fields. Yet overall, Economics, Psychology, and Political Science showed relatively high levels of engagement with OS practices, while Sociology—where qualitative and interpretive methods are more common—lagged significantly behind. This gap suggests that methodological orientation may be a central factor: disciplines that emphasize quantitative, replicable research appear more likely to embrace OS practices, whereas those using more qualitative or interpretivist approaches may adopt them less frequently. Other recent articles have noted adjacent behaviors, such as reproducibility, where sociology lags other disciplines (Moody et al. 2022).

Taking the case of sociology as an example—several potential explanations may account for comparatively low adoption of OS practices in this field. One possibility is institutional: the field has less strong norms or infrastructures that support transparency. Another possibility is methodological: some proportion of methods used in sociology (e.g., ethnography, interviews) may not align easily with OS practices like data sharing or preregistration. However, these explanations may also be connected to deeper epistemological or social commitments. For instance, Weeden (2023) argues that the sociology discipline's relatively slow uptake of transparency norms reflects not just practical or structural barriers, but fundamental disagreement about what constitutes scientific knowledge and how it should be evaluated. Fields differ not only in what methods they use, but in what kinds of knowledge they seek to produce. Disciplinary methods are also embedded within broader judgments about what constitutes legitimate knowledge. In contrast to sociology, Settles et al. (2020) argue that psychology's dominant standards of rigor have historically privileged objectivity and generalizability, researcher neutrality, and quantitative methods, while marginalizing more contextual, constructivist, and participatory approaches. From this perspective, methodological differences may reflect institutionalized epistemological commitments about which questions, methods, and knowers are considered credible. Thus, in more positivist traditions within the social sciences, transparency may be framed as a way to bolster the objectivity and credibility of empirical findings. But in disciplines with stronger interpretivist or humanistic currents, such as anthropology or certain areas of sociology, transparency norms may appear misaligned with core epistemic values like subjectivity, context-dependence, and the intersubjective co-construction of knowledge (Kuhn 1970; Krishnan 2009).

Relatedly, theoretical papers such as Fecher and Friesike (2014) have identified multiple "schools of thought" within the open science movement, each emphasizing different motivations for openness. These schools highlight that while open scholarship may appear unified, the values underpinning it differ across contexts and stakeholders. Indeed, even among open-science advocates, apparent agreement about openness may conceal disagreement about its meaning and goals. In a qualitative study, Grubb and Easterbrook (2011) found limited consensus concerning replication, reproducibility, peer review, and the timing of data and results sharing. Although participants generally endorsed making scientific products accessible, they differed in what they believed should be shared, when sharing should occur, and which problems openness should address. Building on this, Ross-Hellauer et al. (2022) distinguish between two broad rationales in the literature for participating in OS practice: one focused on improving scientific rigor, and another focused on promoting equity and inclusion. These

perspectives are not mutually exclusive; however, they may reflect different assumptions about what science is for, who it serves, and how it should be conducted. Such assumptions likely shape the language, values, and motivations that resonate with researchers across different epistemic and disciplinary communities.

If this is the case, researchers' attitudes towards specific OS practices may depend not only on disciplinary affiliation or methodological training but also on their underlying beliefs about scientific knowledge and scientific communities. Scholars who view science as a means of uncovering objective truths may be especially supportive of practices framed in scientific rigor, verification, or reproducibility. Researchers who perceive scientific communities as exclusionary may place greater weight on accessibility and be especially supportive of practices such as open access. These orientations need not be oppositional; rather, they may provide distinct but complementary foundations for supporting practices. In short, variation in OS attitudes may reflect differences in what researchers believe science is, how credible knowledge is established, and who has access to it. The present study investigates whether these beliefs are associated with support for different components of OS.

To our knowledge, this is the first study to systematically examine whether distinct epistemological and social beliefs help explain why researchers support different components of open science, beyond disciplinary affiliation and methodological practice. Understanding why individual researchers support these different components of OS may inform interventions aimed at increasing OS uptake across and within fields.

## Design

Building on these theoretical accounts and recent empirical findings, the present study investigates whether different epistemological and social beliefs are associated with support for different open science practices across the social sciences. Specifically, we examine whether researchers' positivism and their perceptions of exclusion in science—two dimensions central to debates about the goals of scientific transparency—are associated with attitudes toward particular open science practices. Rather than treating open science support as a single general orientation, we test whether these beliefs show distinct associations with attitudes toward data sharing, preregistration, open access, and replication.

We examined three hypotheses, all specified in advance of data collection. H1 predicted that epistemological and social beliefs would vary across disciplines. H2 predicted that these beliefs would vary by researchers' methodological orientation and their perceptions of the qualitative–quantitative composition of their field. H3 predicted that Positivism and Perceived Exclusion would explain additional variation in practice-specific open science attitudes beyond discipline and methodological orientation.

Prior work has often focused primarily on differences between disciplines or methods. For example, in Ferguson et al. (2023), sociology was found to be the only discipline among the four surveyed that showed substantially lower rates of open science adoption overall. Importantly, it was *also* the only discipline in that sample where a higher proportion of survey respondents reported using qualitative methods. This relatively constrained sample of disciplines made it difficult to determine whether sociology's lower OS engagement reflects institutional differences in infrastructure or norms, the practical constraints of qualitative methods, or deeper epistemological commitments. To clarify these distinctions, the present study expands beyond the original four-discipline framework. We sample researchers across a broader set of social science fields—Anthropology, Business, Communications, Economics, Education, Political Science, Psychology, and Sociology—in order to test whether patterns observed in sociology generalize to other disciplines. This broader and more varied disciplinary sampling allows us to better isolate the relative contributions of field-level norms (discipline),

methodological practices, and epistemological beliefs to researchers' support for and engagement with open science.

## Results

### Sample Characteristics

The final analytic sample consisted of 1,316 researchers drawn from eight social science disciplines, including economics, business, political science, psychology, communications, sociology, education, and anthropology. Of these, 1,298 respondents could be assigned to one of the eight target fields. Sample sizes were relatively balanced across fields, ranging from 150 participants in Business and Economics to 209 in Sociology. The largest disciplinary groups were Sociology ( $n = 209$ ) and Political Science ( $n = 171$ ), followed by Psychology ( $n = 158$ ), Anthropology ( $n = 156$ ), Education ( $n = 153$ ), Communications ( $n = 151$ ), Economics ( $n = 150$ ), and Business ( $n = 150$ ). Relative to the 18,284 unique addresses that received at least one delivered invitation, the final analytic sample represented an overall response rate of 7.2%. Field-specific rates ranged from 4.9% in Business to 11.2% in Sociology (Anthropology, 6.6%; Communications, 5.8%; Economics, 6.8%; Education, 6.6%; Political Science, 8.8%; Psychology, 8.2%).

Participants reported a range of methodological orientations. 749 respondents identified as primarily quantitative, whereas 382 identified as primarily qualitative. Methodological orientation was listed as N/A for 185 respondents.

Reported years of research experience also varied widely. After excluding clearly invalid values, respondents with usable data on this variable ( $n = 1,128$ ) reported an average of 13.73 years of research experience ( $SD = 12.65$ , median = 8, range = 0–65), suggesting substantial variation in career stage across the sample (Fig. 1).

### Sample Characteristics

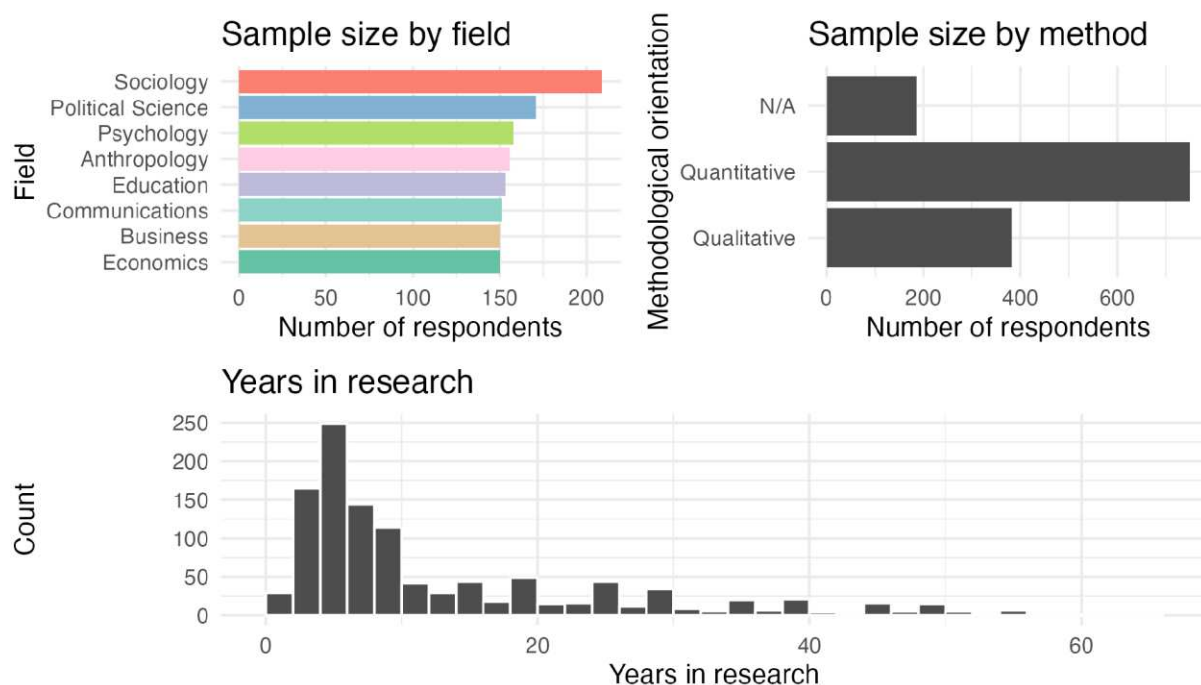


Figure 1 | Sample characteristics. Analytic sample sizes by discipline ( $n = 1,298$  respondents assigned to one of the eight target fields) and by self-reported methodological orientation (quantitative  $n = 749$ , qualitative  $n = 382$ , not reported  $n = 185$ ), and the distribution of years of research experience among respondents with usable values ( $n = 1,128$ ;  $M = 13.73$ ,  $SD = 12.65$ , median = 8, range 0–65).

Scale reliability was evaluated before confirmatory analyses. Positivism showed modest internal consistency ( $\alpha = .688$ ), Perceived Exclusion showed strong internal consistency ( $\alpha = .889$ ), and Neutrality showed weaker internal consistency ( $\alpha = .615$ ). Because Neutrality was less reliable and exploratory, the primary confirmatory models focused on Positivism and Perceived Exclusion. No Positivism item improved internal consistency when removed (alphas from .60 to .67; mean inter-item  $r = .31$ ), indicating that the scale's modest reliability reflects the breadth of the construct rather than a single poorly performing item.

## Variation in Epistemological Beliefs by Discipline

Supporting H1, epistemological and social beliefs varied systematically across disciplines. One-way analyses of variance confirmed significant differences between disciplines on all three scales: Positivism,  $F(7, 1290) = 10.11$ ,  $\eta^2 = .05$ ; Perceived Exclusion,  $F(7, 1290) = 15.82$ ,  $\eta^2 = .08$ ; Neutrality,  $F(7, 1290) = 17.22$ ,  $\eta^2 = .09$ ; all  $p_{\text{BH}} < .001$ . Discipline nonetheless accounted for a modest share of the variance in each belief.

Positivism scores were highest in economics ( $M = 3.15$ ,  $SD = 0.61$ ) and business ( $M = 2.96$ ,  $SD = 0.76$ ), and lowest in anthropology ( $M = 2.61$ ,  $SD = 0.87$ ), education ( $M = 2.61$ ,  $SD = 0.79$ ), and sociology ( $M = 2.64$ ,  $SD = 0.77$ ). Intermediate levels were observed in political science ( $M = 2.87$ ,  $SD = 0.73$ ), psychology ( $M = 2.85$ ,  $SD = 0.75$ ), and communications ( $M = 2.72$ ,  $SD = 0.76$ ). In contrast, Perceived Exclusion scores showed broadly the opposite pattern. Scores were highest in anthropology ( $M = 3.72$ ,  $SD = 0.94$ ) and sociology ( $M = 3.71$ ,  $SD = 0.92$ ), followed by education ( $M = 3.55$ ,  $SD = 1.03$ ), communications ( $M = 3.52$ ,  $SD = 0.98$ ), and psychology ( $M = 3.49$ ,  $SD = 0.96$ ). Lower scores were observed in political science ( $M = 3.35$ ,  $SD = 1.07$ ), economics ( $M = 3.09$ ,  $SD = 1.08$ ), and business ( $M = 2.80$ ,  $SD = 1.05$ ). Neutrality scores broadly mirrored the pattern observed for positivism, with higher values in economics ( $M = 2.55$ ,  $SD = 0.59$ ) and business ( $M = 2.48$ ,  $SD = 0.63$ ), and lower values in sociology ( $M = 1.97$ ,  $SD = 0.69$ ) and anthropology ( $M = 1.97$ ,  $SD = 0.72$ ). However, variability across disciplines was more limited, and interpretation of this scale should be treated with caution given its lower reliability.

Taken together, these patterns indicate that epistemological and social beliefs are not evenly distributed across the social sciences, but instead align with well-known disciplinary differences in methodological traditions and theoretical orientation (Fig. 2).

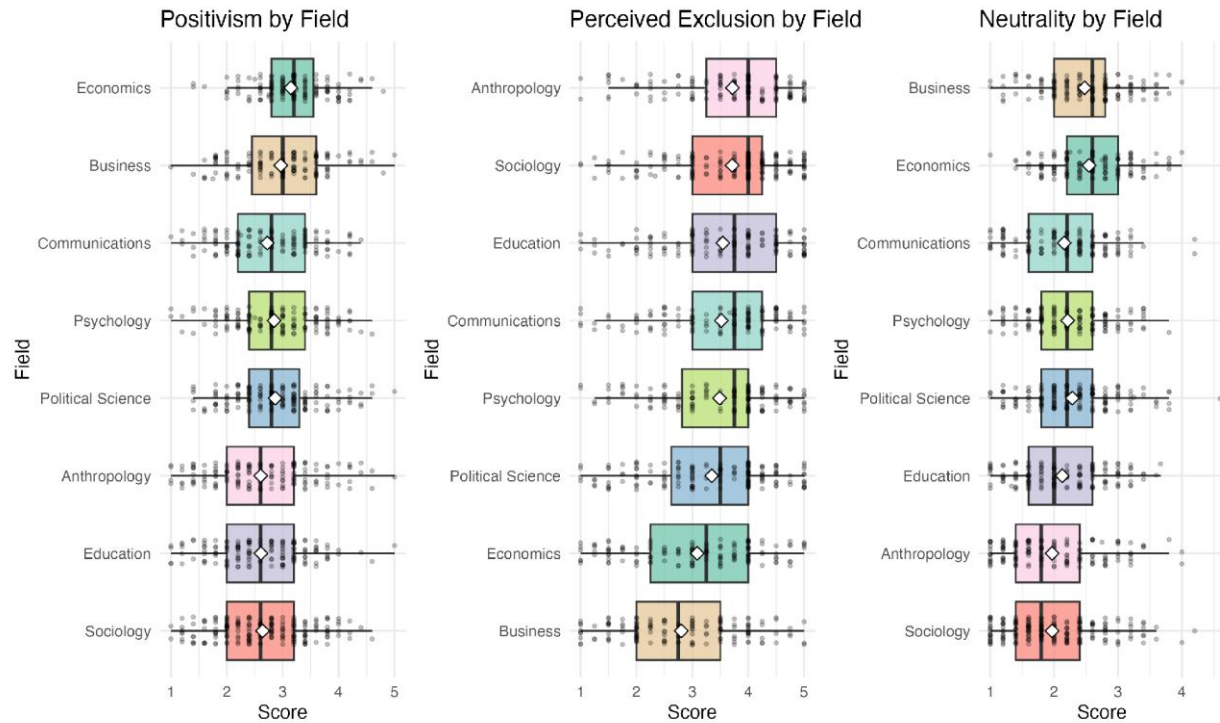


Figure 2 | Epistemological and social beliefs by discipline. Distributions of Positivism, Perceived Exclusion and Neutrality scores across the eight target disciplines ( $n = 1,298$  respondents assigned to a target field). Boxes show medians and interquartile ranges, points show individual respondents, and white diamonds show means. Neutrality was exploratory and had lower internal consistency than the other scales.

### Variation in Epistemological Beliefs by Methodological Orientation

Supporting H2, epistemological and social beliefs varied with researchers' self-reported methodological orientation and with their perceptions of their field's methodological composition.

Researchers identifying as primarily quantitative reported higher levels of positivism ( $M = 2.97$ ,  $SD = 0.72$ ) than those identifying as primarily qualitative ( $M = 2.48$ ,  $SD = 0.80$ ). In contrast, qualitative researchers reported higher Perceived Exclusion scores ( $M = 3.68$ ,  $SD = 0.97$ ) than quantitative researchers ( $M = 3.27$ ,  $SD = 1.05$ ). Neutrality scores followed a pattern similar to that of positivism, with higher values among quantitative researchers than qualitative researchers (Fig. 3). These differences between methodological orientations were statistically significant for all three scales: Positivism,  $F(1, 1129) = 107.94$ ,  $d = 0.65$ ; Perceived Exclusion,  $F(1, 1129) = 40.37$ ,  $d = -0.40$ ; Neutrality,  $F(1, 1129) = 114.22$ ,  $d = 0.67$ ; all  $p_{BH} < .001$ .

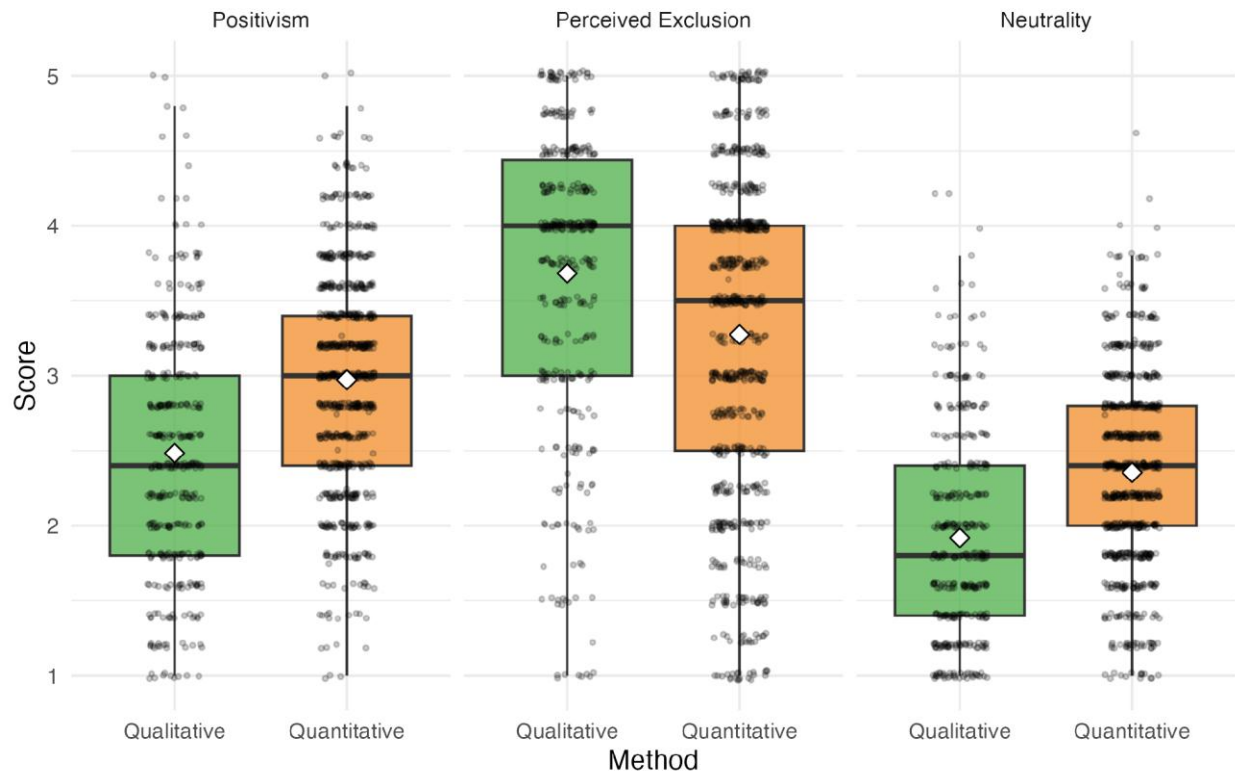


Figure 3 | Epistemological and social beliefs by methodological orientation. Distributions of Positivism, Perceived Exclusion and Neutrality scores for researchers who described their work as primarily quantitative ( $n = 749$ ) or primarily qualitative ( $n = 382$ ). Boxes show medians and interquartile ranges, points show individual respondents, and white diamonds show means. Quantitative researchers scored higher on Positivism and Neutrality and lower on Perceived Exclusion (Positivism,  $F(1, 1129) = 107.94$ ,  $d = 0.65$ ; Perceived Exclusion,  $F(1, 1129) = 40.37$ ,  $d = -0.40$ ; Neutrality,  $F(1, 1129) = 114.22$ ,  $d = 0.67$ ; all  $p_{\text{BH}} < .001$ ).

To examine whether epistemological beliefs varied with perceptions of disciplinary composition, we analyzed associations between participants' beliefs and their estimates of the proportion of qualitative research in their field.

Higher perceived proportions of qualitative research were associated with lower positivism scores ( $r = -.14$ , 95% CI  $[-.20, -.09]$ ,  $p < .001$ ) and lower neutrality scores ( $r = -.13$ , 95% CI  $[-.18, -.07]$ ,  $p < .001$ ). That is, participants who viewed their field as having more qualitative researchers proportionally tended to report weaker endorsement of positivist and neutrality-oriented beliefs. In contrast, perceived qualitative composition was positively associated with Perceived Exclusion ( $r = .10$ , 95% CI  $[.04, .15]$ ,  $p = .001$ ), such that participants who perceived their field as more qualitative reported stronger perceptions of exclusion.

Although these associations were modest in magnitude, they were directionally consistent with patterns observed across disciplines and methodological self-identification (Fig. 4). Taken together, these results suggest that epistemological beliefs are associated not only with researchers' disciplinary affiliation and methodological orientation, but also with their perceptions of the broader methodological composition of their field.

## Epistemological and Social Beliefs by Perceived Disciplinary Method

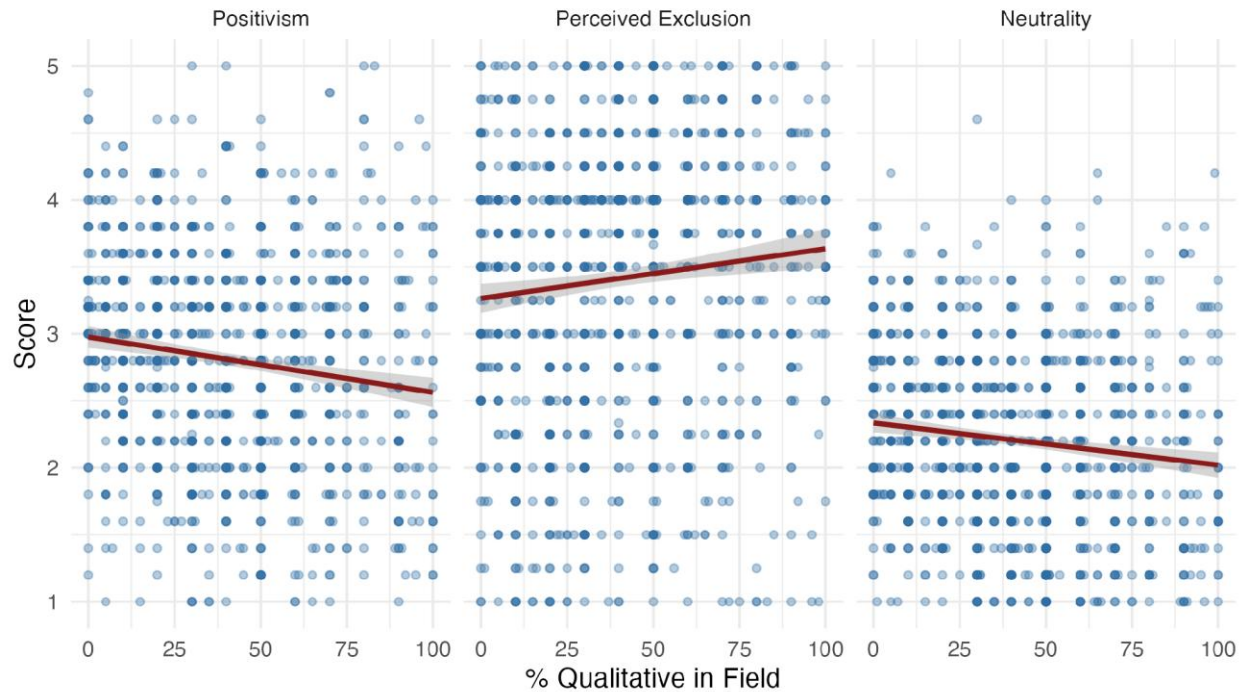


Figure 4 | Belief scores by the perceived qualitative composition of the respondent's discipline. Each point is one respondent's belief score plotted against their own estimate of the percentage of research in their field that is qualitative ( $n = 1,138$  respondents with both measures). Lines and shaded bands show ordinary least-squares fits with 95% confidence intervals. Higher perceived qualitative composition was associated with lower Positivism ( $r = -.14$ , 95% CI  $[-.20, -.09]$ ,  $p < .001$ ) and lower Neutrality ( $r = -.13$ , 95% CI  $[-.18, -.07]$ ,  $p < .001$ ), and with higher Perceived Exclusion ( $r = .10$ , 95% CI  $[.04, .15]$ ,  $p = .001$ ).

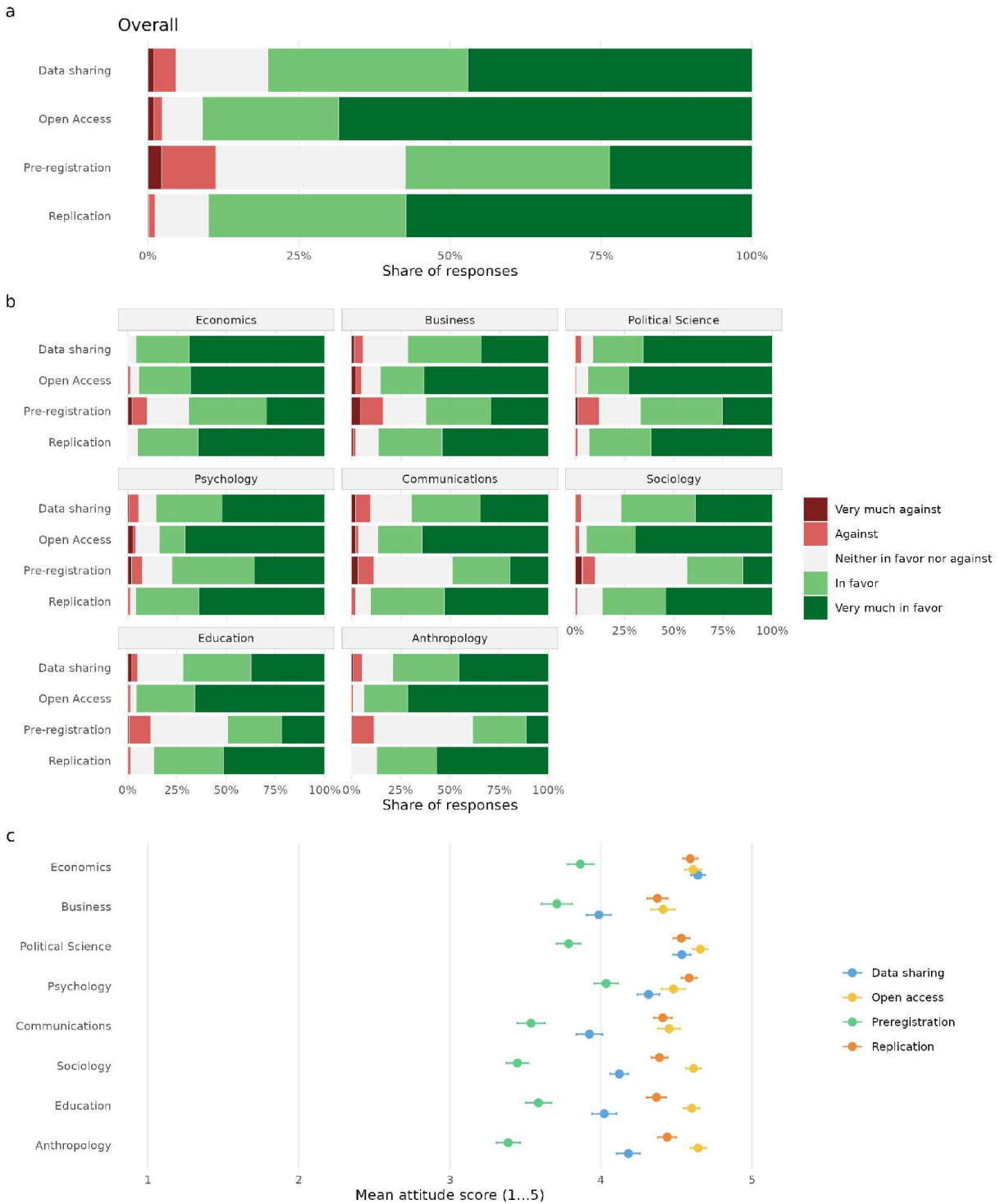
## Descriptive Patterns in Open Science Attitudes

Across domains, participants expressed consistently positive attitudes toward open science practices. Mean ratings exceeded the midpoint of the scale across all fields and domains. Attitudes were highest for open access ( $M_s = 4.41\text{--}4.66$ ) and replication ( $M_s = 4.37\text{--}4.59$ ), followed by data sharing ( $M_s = 3.92\text{--}4.64$ ). In contrast, preregistration received comparatively lower endorsement ( $M_s = 3.39\text{--}4.03$ ), though still above the scale midpoint. This pattern held across all disciplines, with relatively modest variation between fields (Fig. 5). Responses were strongly concentrated at the upper end of each scale: 80% of respondents were in favor or very much in favor of data sharing, 91% of open access, 90% of replication and 57% of preregistration, and fewer than 5% were against or very much against any practice except preregistration (11%). The models reported below therefore distinguish mainly between degrees of support rather than between support and opposition.

Preregistration was the least strongly endorsed open science practice across fields, although mean ratings remained above the scale midpoint in all disciplines. Support was highest in Psychology ( $M = 4.03$ ), followed by Economics ( $M = 3.86$ ), Political Science ( $M = 3.79$ ), and Business ( $M = 3.71$ ). It was lowest in Anthropology ( $M = 3.39$ ), Sociology ( $M = 3.45$ ), Communications ( $M = 3.54$ ), and Education ( $M = 3.59$ ). Thus, disciplinary differences were most

visible for data sharing and preregistration, whereas open access and replication were more uniformly supported across the social sciences.

For each practice, respondents additionally estimated the percentage of researchers in their discipline who were very much in favor, in favor, neutral, against, or very much against the practice. We operationalized perceived support as the estimated percentage of researchers in the respondent's field who were either in favor or very much in favor. Perceived support for open access ranged from approximately 50% in Business to 64% in Political Science, Anthropology, and Economics. Perceived support for data sharing was highest in Economics (approximately 75%) and Political Science (approximately 65%), followed by Psychology (approximately 57%); it was lowest in Communications (approximately 41%). Perceived support for preregistration showed the greatest field variation, ranging from approximately 19% in Anthropology and 27% in Sociology to 50% in Psychology. Perceived support for replication was highest in Economics (approximately 60%), Political Science (approximately 58%), and Psychology (approximately 56%), and lowest in Anthropology (approximately 35%). These perceived-support estimates should be interpreted as exploratory because they asked respondents to approximate the distribution of attitudes among others in their field (Supplementary Fig. 1).



**Figure 5 | Open science attitudes by discipline.** Response distributions for data sharing, open access, preregistration and replication are shown overall and separately by discipline. Categories range from very much against to very much in favor; percentages are calculated among respondents with a valid response for the corresponding item.

## Model Comparisons

To examine whether epistemological and social beliefs were associated with open science attitudes above and beyond disciplinary and methodological differences, we conducted nested ordinal logistic regressions for the four attitude outcomes. For each outcome, Model 1 included field, Model 2 added methodological orientation (qualitative or quantitative), and Model 3 added Positivism and Perceived Exclusion. We compared models using likelihood-ratio tests and Akaike information criterion (AIC). This nested sequence, and the designation of Positivism and Perceived Exclusion as the focal belief terms, were specified in the study protocol before data collection (see Analysis plan).

Supporting H3, adding Positivism and Perceived Exclusion improved ordinal model fit beyond both field and methodological orientation for all four attitudes. For data sharing attitudes, Model 3 improved over Model 2 ( $p_{\text{BH}} < .001$ ; AIC = 2551.10, 2512.30, and 2500.13 for Models 1–3, respectively). For preregistration attitudes, Model 3 improved over Model 2 ( $p_{\text{BH}} = .008$ ; AIC = 2753.08, 2738.70, and 2732.92). For open access attitudes, Model 3 improved over Model 2 ( $p_{\text{BH}} < .001$ ; AIC = 1938.36, 1939.52, and 1898.76), although methodological orientation alone did not improve over field alone ( $p = .360$ ). For replication attitudes, Model 3 improved over Model 2 ( $p_{\text{BH}} < .001$ ; AIC = 2109.17, 2053.80, and 2040.54).

Across outcomes, AIC was lowest for Model 3, indicating better fit when epistemological and social beliefs were included. The AIC reductions were largest for replication and data sharing, followed by open access and preregistration (Fig. 6).

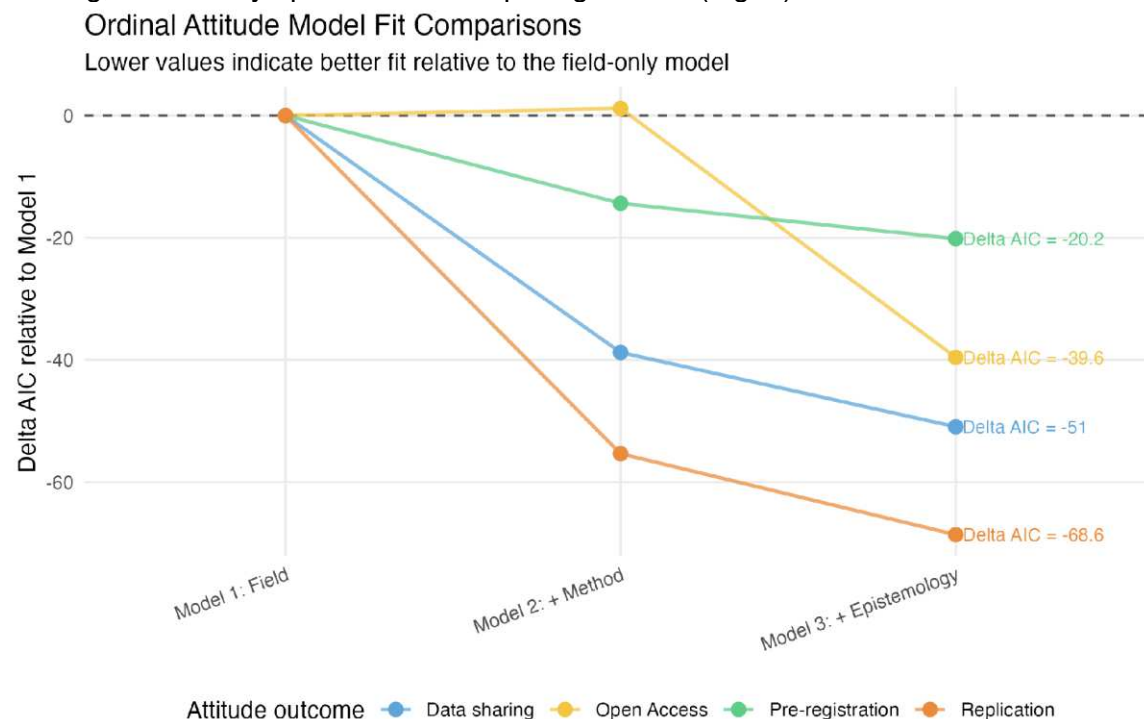


Figure 6 | Comparison of ordinal attitude models. Change in Akaike information criterion (AIC) relative to the field-only model for models adding methodological orientation and then Positivism and Perceived Exclusion. Lower values indicate better relative fit. Models were estimated on complete cases for each outcome.

To estimate the specific associations of the focal beliefs, we inspected coefficients from the full models, which included field, methodological orientation, Positivism, and Perceived Exclusion. Full model coefficients for all terms are reported in Table 1.

Positivism was significantly associated with attitudes toward all four open science practices in the ordinal models, although the direction differed across practices. A one-unit increase in Positivism was associated with higher odds of stronger support for data sharing (OR = 1.33, 95% CI [1.14, 1.57],  $p_{\text{BH}} = .001$ ), preregistration (OR = 1.29, 95% CI [1.10, 1.51],  $p_{\text{BH}} = .004$ ), and replication (OR = 1.43, 95% CI [1.21, 1.69],  $p_{\text{BH}} < .001$ ). Positivism was negatively associated with open access attitudes (OR = 0.80, 95% CI [0.67, 0.96],  $p_{\text{BH}} = .022$ ), indicating lower odds of stronger open access support after accounting for field and method. The negative open-access association was sensitive to model specification. In the prespecified model, which additionally adjusted for perceived disciplinary composition, years of research, strength of researcher identity and gender, it attenuated to OR = 0.87, 95% CI [0.72, 1.06] (Supplementary Table 2). Years of research accounted for this attenuation: it was associated with both higher Positivism ( $r = .20$ , 95% CI [.14, .26]) and weaker open-access support ( $r = -.27$ , 95% CI [-.33, -.22]), so more experienced researchers were both more positivist and less supportive of open access. The seven other belief–attitude associations were unchanged under that specification, and the Perceived Exclusion associations with data sharing and open access strengthened slightly.

Perceived Exclusion showed a distinct pattern. Higher Perceived Exclusion was associated with stronger support for data sharing (OR = 1.18, 95% CI [1.05, 1.33],  $p_{\text{BH}} = .009$ ) and open access (OR = 1.43, 95% CI [1.26, 1.64],  $p_{\text{BH}} < .001$ ), but was not significantly associated with preregistration or replication attitudes after covariate adjustment.

Thus, Positivism was associated with attitudes toward every practice, positively for data sharing, preregistration, and replication but negatively for open access, whereas Perceived Exclusion was positively associated with data-sharing and open-access attitudes (Fig. 7).

Table 1 | Ordinal logistic models of attitudes toward four open science practices. Entries are odds ratios with 95% confidence intervals from the full model for each attitude outcome, which includes discipline, methodological orientation, Positivism and Perceived Exclusion. Odds ratios above 1 indicate greater odds of stronger support. Economics is the reference discipline and primarily qualitative work the reference methodological orientation. Asterisks mark coefficients whose 95% confidence interval excludes 1; Benjamini–Hochberg-adjusted  $p$  values for the focal belief coefficients are reported in the text. Models were estimated on complete cases.

| Term                                     | Data sharing       | Open access       | Preregistration    | Replication       |
|------------------------------------------|--------------------|-------------------|--------------------|-------------------|
| <b>Discipline (reference: Economics)</b> |                    |                   |                    |                   |
| Business                                 | 0.26 [0.16, 0.42]* | 0.74 [0.44, 1.24] | 0.89 [0.56, 1.43]  | 0.81 [0.49, 1.32] |
| Political Science                        | 0.98 [0.59, 1.63]  | 1.02 [0.60, 1.72] | 0.95 [0.61, 1.49]  | 1.23 [0.75, 2.02] |
| Psychology                               | 0.48 [0.29, 0.77]* | 0.77 [0.46, 1.31] | 1.52 [0.97, 2.38]  | 1.14 [0.69, 1.86] |
| Communications                           | 0.28 [0.17, 0.45]* | 0.60 [0.35, 1.01] | 0.60 [0.37, 0.96]* | 1.04 [0.63, 1.71] |

| Term                                                       | Data sharing       | Open access        | Preregistration    | Replication        |
|------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|
| Sociology                                                  | 0.40 [0.25, 0.64]* | 0.86 [0.52, 1.43]  | 0.54 [0.35, 0.84]* | 1.06 [0.66, 1.71]  |
| Education                                                  | 0.34 [0.20, 0.56]* | 0.75 [0.44, 1.27]  | 0.73 [0.45, 1.19]  | 1.03 [0.62, 1.71]  |
| Anthropology                                               | 0.52 [0.31, 0.87]* | 0.92 [0.52, 1.61]  | 0.50 [0.31, 0.81]* | 1.39 [0.82, 2.36]  |
| <b>Methodological orientation (reference: qualitative)</b> |                    |                    |                    |                    |
| Quantitative                                               | 2.13 [1.63, 2.77]* | 1.07 [0.79, 1.45]  | 1.59 [1.21, 2.08]* | 2.51 [1.90, 3.31]* |
| <b>Epistemological and social beliefs</b>                  |                    |                    |                    |                    |
| Positivism                                                 | 1.33 [1.14, 1.57]* | 0.80 [0.67, 0.96]* | 1.29 [1.10, 1.51]* | 1.43 [1.21, 1.69]* |
| Perceived Exclusion                                        | 1.18 [1.05, 1.33]* | 1.43 [1.26, 1.64]* | 1.08 [0.96, 1.22]  | 1.08 [0.95, 1.22]  |
| <b>n</b>                                                   | <b>1,112</b>       | <b>1,114</b>       | <b>1,018</b>       | <b>1,109</b>       |

As a robustness check, we refit the attitude models using ordinary linear regression on the 1–5 scores. The linear and ordinal models produced the same coefficient directions and the same statistical-significance conclusions for all eight Positivism- and Perceived-Exclusion-by-attitude associations. Adding epistemological and social beliefs also improved model fit for all four attitudes under both approaches.

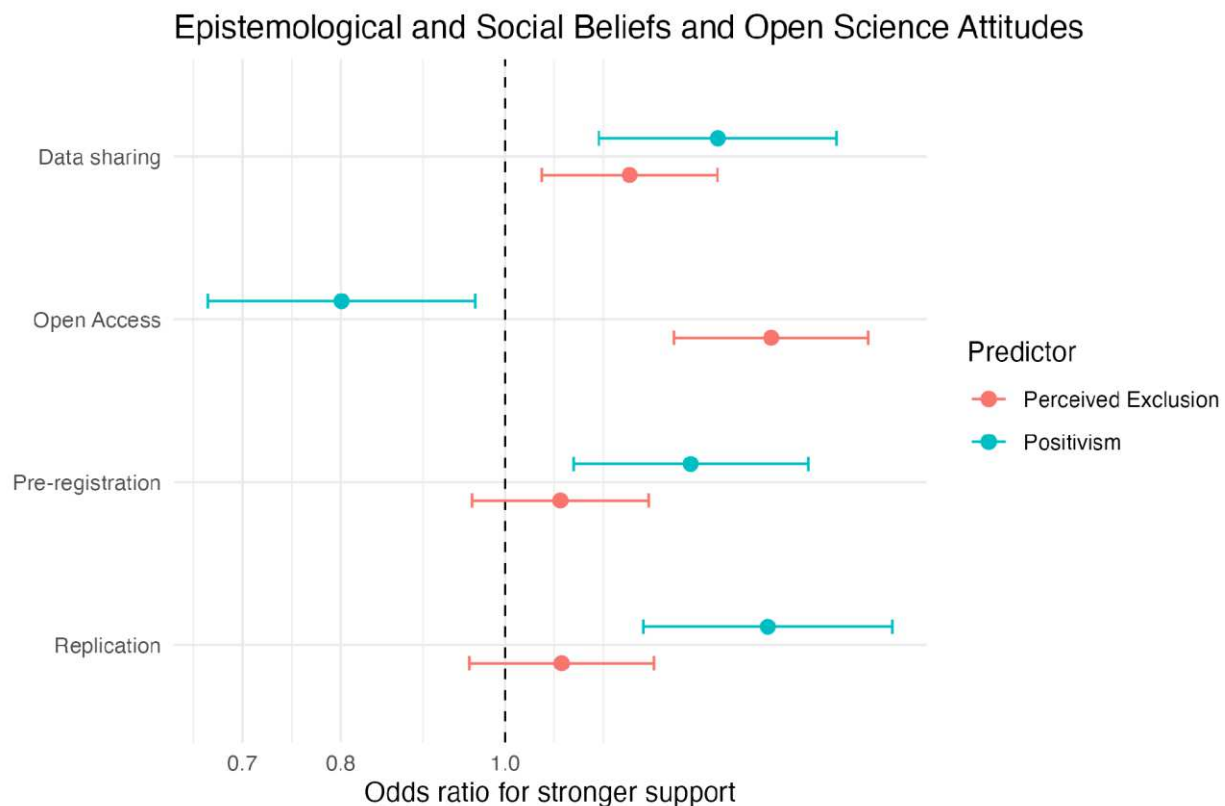


Figure 7 | Epistemological and social beliefs and open science attitudes. Points are odds ratios and horizontal lines are 95% confidence intervals from the full ordinal logistic models, adjusted for discipline and methodological orientation. Values above 1 indicate greater odds of stronger support; the dashed line denotes an odds ratio of 1. Model sample sizes were 1,112 for data sharing, 1,114 for open access, 1,018 for preregistration and 1,109 for replication.

## Discussion

The present study examined whether variation in open science (OS) attitudes across the social sciences is associated with individual researchers' epistemological and social beliefs above and beyond discipline and method. Across a large and diverse sample within the social sciences, three central findings emerged. First, epistemological and social beliefs varied systematically across disciplines and methodological orientations. Second, these beliefs explained additional variation in specific open science attitudes beyond discipline and method. Third, they corresponded to different patterns of support: positivism was associated with attitudes toward all four practices, positively for preregistration, replication, and data sharing but negatively for open access, whereas perceived exclusion was associated positively with open access and data sharing.

These beliefs were associated with open science attitudes even after adjustment for discipline and methodological orientation. Disciplinary differences therefore were not reducible to methodological orientation alone: researchers within the same field and methodological category still differed in the scientific goals they prioritized and in the forms of openness they endorsed. Epistemological and social beliefs may thus represent an individual-level source of variation alongside structural constraints associated with discipline. This study extends previous survey work in the social sciences (Ferguson et al. 2023) by examining a broader set of fields and beliefs about both scientific knowledge and scientific communities. Because the hypotheses and the analysis plan were fixed before data collection, these three tests are confirmatory rather than post hoc; the analyses of perceived support, Neutrality and years in research are identified as exploratory throughout.

A central contribution of this study is its treatment of open science not as a single orientation but as a collection of practices that may be valued for different reasons. Researchers' epistemic and social commitments were associated with distinct patterns of support, providing individual-level evidence for conceptual accounts of multiple open science "schools of thought" grounded in different assumptions about the goals and values of scientific practice (Fecher and Friesike 2014).

The present findings align with broader evidence that movements to reform science are organized around multiple meanings and values. Even among advocates of openness, researchers disagree about the goals of replication, sharing, and accessibility and about how those goals should be implemented (Grubb and Easterbrook 2011). Moreover, reform movements may develop distinct rhetorical cultures: for example, open-science scholarship that uses more communal and prosocial language than reproducibility scholarship included greater representation of women in high-status authorship positions (Murphy et al. 2020).

### Implications for promoting open science

These findings may inform efforts to increase support for open science practices. Existing interventions operate at structural and individual levels and can invoke different norms and motivations. Perceived norms are central to accounts of scientific culture change, including the Pyramid of Culture Change and Easy, Attractive, Social, and Timely frameworks (Nosek 2019; Robson et al. 2021; Nosek et al. 2022).

For example, the Center for Open Science's 'Pyramid of Culture Change' conceptualizes research as a social system and proposes coordinated interventions that make open practices

possible, easy, normative, rewarding, and required. Within this framework, practices become normative when early adopters make them visible and aspirational within their communities; requirements represent a later and distinct form of change implemented through journal, funder, and institutional policies.

The present findings help specify an additional dimension of this normative process. A practice can become normative because researchers observe others performing it, because institutions communicate that it is expected, or because the practice is understood to advance a valued scientific goal. Our results speak most directly to this third pathway: what might be called the practice's normative appeal. Researchers may agree that open science is desirable while differing in the reasons they regard particular practices as worthwhile. Appeals to rigor, verification, and reproducibility may align more closely with positivist beliefs, whereas appeals to accessibility, inclusion, and collective benefit may align more closely with the perception that science is currently exclusionary.

As a result, uniform messaging about the benefits of open science may not be equally effective across communities. Appeals to rigor and reproducibility may resonate more strongly with researchers holding positivist beliefs, whereas appeals to equity and accessibility may align more closely with perceptions of exclusion in science. This does not imply that researchers belong to mutually exclusive epistemic audiences or that each practice should be associated with only one rationale. Instead, open science practices can often be connected to multiple scientific values. Data sharing, for example, can facilitate both verification and more equitable access to scientific resources. Presenting complementary rationales may therefore help communities see open practices as relevant to their own scientific commitments. Structural interventions remain essential, but efforts to make open science possible, easy, rewarding, or required may be more successful when researchers also understand why those practices serve the purposes they value.

## Limitations

First, this observational study relied on cross-sectional survey data and therefore cannot establish that epistemological or social beliefs cause open science attitudes. Unmeasured characteristics could be associated with both the focal beliefs and attitudes, and the direction of association may also be reciprocal. Future work could test causal mechanisms by experimentally varying the rationale used to frame an open science practice and measuring subsequent attitudes or behavioral intentions.

Second, the measures may not have fully captured the intended constructs. Although the Perceived Exclusion scale demonstrated strong internal consistency, its items assessed only whether women and racialized groups are represented and welcomed in science. Higher scores therefore indicate that a respondent perceives scientific communities as exclusionary, which is distinct from a normative commitment to epistemic inclusion. Disciplinary differences in Perceived Exclusion scores also may not reflect objective demographic composition; prior research, for example, suggests lower demographic diversity in economics than in sociology (Lockhart et al. 2025; National Center for Science and Engineering Statistics 2023). The present study cannot separate descriptive perceptions from evaluative commitments. Future research should use more direct measures of epistemic inclusion, including beliefs about whether diverse perspectives improve knowledge, whose knowledge is treated as credible, and whether accessibility and representation are central goals of science.

Finally, the primary outcomes assessed in this paper were attitudes toward specific open science practices rather than behavioral adoption. However, attitudes do not necessarily translate directly into behavior. Whether researchers actually preregister studies, share data, publish open access, or conduct replications may depend on additional factors, including journal and funder requirements, collaborator expectations, available infrastructure, data sensitivity, project design, and field-specific norms. Future research should examine how epistemological beliefs and open science attitudes relate to behavioral adoption, both generally and over time,

including whether attitudes mediate the relationship between epistemological commitments and actual open science practice. The high overall level of support also limited variability in the outcomes, so these models speak to predictors of the intensity of support rather than of outright opposition.

### Conclusion

Taken together, the present findings suggest that support for open science reflects not only disciplinary norms or methodological constraints, but also researchers' beliefs about the goals and values of science. Different open science practices may appeal to distinct, though overlapping, epistemic commitments: practices associated with rigor and reproducibility may resonate with positivist beliefs, whereas practices associated with accessibility may align more closely with perceptions of exclusion in science. Recognizing this plurality can provide a more nuanced account of variation in open science attitudes and inform efforts to communicate the value of openness across diverse research communities. Open science may therefore be most effectively understood not as a single, uniform movement, but as a collection of practices supported through multiple pathways and for multiple scientific purposes.

## Methods

### Sampling Design

To ensure a diverse sample of researchers across disciplines, this study used a stratified sampling approach by discipline. The goal was to achieve approximately equal numbers of participants across the eight target disciplines (anthropology, business, communications, economics, education, political science, psychology, and sociology) to allow for meaningful between-group comparisons. A target of at least 150 respondents was planned per discipline, for a minimum total sample of approximately 1,200 participants.

An a priori power analysis was conducted using the pwr package in R. Assuming a small effect size (Cohen's  $f^2 = 0.02$ ), a significance threshold of  $\alpha = 0.05$ , power = 0.95, and a regression model with eight predictors, the required total sample size is approximately 1,145 participants. This estimate is conservative and consistent with prior research on individual predictors of open science engagement (e.g., Ferguson et al. 2023). It ensures adequate statistical power to detect modest but theoretically meaningful effects of epistemological beliefs on support for open science practices.

### SAMPLING FRAME

The primary sampling frame consisted of the top 292 U.S. universities with social science graduate programs, as identified from U.S. News and World Report rankings (2022; see also Frishberg 2025 for archived ranking data). We used the rvest package (Wickham 2025) in R (R Core Team 2026) to collect publicly available email addresses of social science scholars in anthropology, business, communications, economics, education, political science, psychology, and sociology. Survey invitations were sent to randomly selected scholars within the disciplinary sampling strata. Across all recruitment waves, invitation attempts were sent to 18,552 unique email addresses. Of these, 18,284 received at least one delivered invitation; 268 addresses had only failed-delivery or hard-bounce statuses.

### SURVEY ADMINISTRATION

Survey invitations included reminder emails after initial contact. Participants completed the survey online and provided informed consent before participation. This study was reviewed

and approved as exempt under Category 2 by the institutional review board of the authors' institution. The reviewing institution and protocol number are withheld here to preserve double-anonymized review and will be supplied on acceptance. Recruitment email addresses were used only for survey administration and were not included in the analytic dataset.

## Scale Construction and Reliability Items

Epistemological beliefs were measured using adapted items from existing science-belief scales. Positivism was measured with five items adapted from the science positivism scale developed by Steel et al. (2004). These items assessed the extent to which respondents endorsed the view that scientific knowledge is objective, cumulative, and best established through systematic empirical methods. Example items included beliefs that science provides objective knowledge about the world and that the scientific method is the only certain way to determine what is true or false.

Perceived Exclusion was measured with four items adapted from Cobern's (2001) Thinking about Science Survey Instrument (TSSI). These items assessed respondents' perceptions of whether women and racialized minorities are equally welcomed within scientific communities. Two items were reverse-coded so that higher values indicated a stronger perception that scientific communities are exclusionary. We refer to this scale as Perceived Exclusion rather than Inclusivity because higher scores index the perception that science is currently unwelcoming to women and racialized minorities.

We also included an exploratory Neutrality scale assessing beliefs about whether scientific questions and conclusions can be separated from social and political context. This scale was treated as exploratory as it was developed for the present study rather than taken directly from a previously validated measure. Two of the five Neutrality items (items 1 and 3)

were reverse-coded so that higher scores indicated stronger endorsement of the view that scientific questions and conclusions can be separated from social and political context.

## Epistemological belief items

### **Positivism (modified from the science positivism scale of Steel et al. 2004)**

1. Use of the scientific method is the only certain way to determine what is true or false about the world
2. The advance of knowledge is a linear process driven by key experiments
3. Science provides objective knowledge about the world
4. Facts describe true states of affairs about the world
5. For most scientific questions, there is only one correct answer

### **Perceived Exclusion (modified from the Thinking about Science Survey Instrument, TSSI; Cobern 2001)**

1. The scientific community is mostly dominated by white men and is often unfriendly to minority people
2. The scientific community is mostly dominated by men and is often unfriendly to women
3. In the reality of the community, women are welcome in science just as much as men are (reverse coded)
4. In the reality of the community, African Americans and other minority people are just as welcome in the scientific community as white people are (reverse coded)

### **Science Neutrality (exploratory)**

1. The questions researchers choose to study are influenced by social and political contexts.
2. No matter the topic, researchers can make objective conclusions if they follow the scientific method strictly.
3. The idea of “complete neutrality” in science is unrealistic.
4. Research produced by people with a personal stake in the topic is less reliable than research from people without such a stake.
5. Scientists who openly advocate for a cause cannot produce trustworthy research on that cause.

### **OSS Items**

Attitudes toward data sharing, preregistration, open access, and replication were assessed as four separate ordinal outcomes rather than combined into a single scale. For each practice, respondents indicated support using five ordered categories from Very much against to Very much in favor. Separate outcomes preserve the practice-specific differences central to H3; accordingly, internal-consistency reliability is not applicable to a composite attitude scale. Each module opened with a definition of the practice and then asked a single attitude item: “What is your opinion of sharing data?”, “What is your opinion of preregistration?”, “What is your opinion of open access publishing?” and “What is your opinion of replication studies?” Response options were Very much against, Against, Neither in favor nor against, In favor, and Very much in favor. Perceived support was measured with a companion item for each practice: “In your estimation, what is the distribution of opinion across your discipline about [practice]? In other words, what percentage of researchers fall into each opinion category listed below?”, with

respondents distributing percentages across the same five categories. Full item and definition wording is reproduced in Supplementary Table [X].

1. Module: Data Sharing Fundamentals — Attitudes and Perceived Norms
2. Module: Preregistration Fundamentals — Attitudes and Perceived Norms
3. Module: Open Access Fundamentals — Attitudes and Perceived Norms
4. Module: Replication Fundamentals — Attitudes and Perceived Norms

#### **Demographics**

1. How many years have you been doing research or scholarship in [participant field]? (For example, if you are a graduate student and are in year 3, and you were a research assistant for 2 years, you would put '5')
2. How much do you identify (i.e., it's an important part of 'who you are') with being a researcher in [participant field]?
3. What discipline did you get your PhD in?
4. What type of research do you primarily conduct?
  - a. Qualitative / Quantitative
5. Please describe the method you typically use in your research e.g., experimental design, big data, ethnography etc
  - a. Open response
6. What would you say is the proportion of Qualitative to Quantitative research in your discipline?
  - a. 1-100 scale

## **Analysis plan**

The hypotheses, measures, sampling design and analysis plan were prespecified in a dated study protocol finalized on 11 November 2025, approximately four weeks before data collection began; the three hypotheses reported here were fixed at that point. The study was not preregistered in a public registry. That version of the protocol is deposited, unmodified, at <https://doi.org/10.5281/zenodo.22901826>. The protocol was followed in its sampling frame, power analysis, belief measures and model structure. Deviations from it, together with the prespecified outcome-neutral checks and their results, are itemised in Supplementary Note 1.

All analyses were conducted in R. Descriptive analyses summarized discipline, methodological orientation, years in research, the belief scales, scale reliability and open science attitudes. Hypotheses H1 and H2 were tested with one-way analyses of variance and accompanying effect sizes, and the association between beliefs and perceived disciplinary composition with Pearson correlations. For H3, the primary confirmatory analyses used ordinal logistic regression for each five-category attitude outcome, comparing a field-only model, a model adding methodological orientation, and a full model adding Positivism and Perceived Exclusion by likelihood-ratio tests and AIC. For the focal belief coefficients we report odds ratios, 95% confidence intervals and Benjamini–Hochberg-adjusted p values within each family of related tests; exploratory analyses are reported unadjusted. Three robustness checks were run: ordinary linear regression on the 1–5 attitude scores, which produced the same coefficient directions and conclusions; an assessment of the proportional-odds assumption (Supplementary Table 1); and the prespecified specification adding perceived disciplinary composition, years of research, researcher identity and gender as covariates (Supplementary Table 2). Analyses of perceived support, Neutrality and years in research were exploratory.

## Data and code availability

The de-identified analytic dataset, codebook and analyses supporting the findings of this study are available at <https://doi.org/10.5281/zenodo.22901826>.

## References

- Anderson, Richard G., William H. Greene, B. D. McCullough, and H. D. Vinod. 2008. "The Role of Data/Code Archives in the Future of Economic Research." *Journal of Economic Methodology* 15 (1): 99–119. <https://doi.org/10.1080/13501780801915574>.
- Bisco, Ralph L. 1966. "Social Science Data Archives: A Review of Developments." *American Political Science Review* 60 (1): 93–109. <https://doi.org/10.2307/1953810>.
- Cobern, William. 2001. "The Thinking about Science Survey Instrument (TSSI): An Instrument for the Quantitative Study of Socio-Cultural Sources of Support and Resistance to Science." Scientific Literacy and Cultural Studies Project. Western Michigan University. [https://scholarworks.wmich.edu/science\\_slcsp/37](https://scholarworks.wmich.edu/science_slcsp/37).
- Desrosières, Alain. 2002. *The Politics of Large Numbers: A History of Statistical Reasoning*. Translated by Camille Naish. Cambridge, MA: Harvard University Press.
- Fecher, Benedikt, and Sascha Friesike. 2014. "Open Science: One Term, Five Schools of Thought." In *Opening Science*, edited by Sönke Bartling and Sascha Friesike. Springer International Publishing. [https://doi.org/10.1007/978-3-319-00026-8\\_2](https://doi.org/10.1007/978-3-319-00026-8_2).
- Ferguson, Joel, Rebecca Littman, Garret Christensen, et al. 2023. "Survey of Open Science Practices and Attitudes in the Social Sciences." *Nature Communications* 14 (1): 5401. <https://doi.org/10.1038/s41467-023-41111-1>.
- Frishberg, Aron. 2025. Archive-of-US-News-College-Rankings. Computer software. GitHub. <https://github.com/frishberg/Archive-of-US-News-College-Rankings>.
- Grubb, Alicia M., and Steve M. Easterbrook. 2011. "On the Lack of Consensus over the Meaning of Openness: An Empirical Study." *PLoS ONE* 6 (8): e23420. <https://doi.org/10.1371/journal.pone.0023420>.
- King, Gary. 1995. "Replication, Replication." *PS: Political Science & Politics* 28 (3): 444–52. <https://doi.org/10.2307/420301>.
- Krishnan, Armin. 2009. "What Are Academic Disciplines? Some Observations on the Disciplinarity vs. Interdisciplinarity Debate." National Centre for Research Methods. [https://eprints.ncrm.ac.uk/id/eprint/783/1/what\\_are\\_academic\\_disciplines.pdf](https://eprints.ncrm.ac.uk/id/eprint/783/1/what_are_academic_disciplines.pdf).
- Kuhn, Thomas S. 1970. *The Structure of Scientific Revolutions*. 2nd ed., enlarged. Chicago: University of Chicago Press.

Lockhart, Jeffrey W., Molly M. King, and Christin Munsch. 2025. "Who Authors Social Science? Demographics and the Production of Knowledge." *Social Currents* 12 (1): 17–29. <https://doi.org/10.1177/23294965241246805>.

Moody, James W., Lisa A. Keister, and Maria C. Ramos. 2022. "Reproducibility in the Social Sciences." *Annual Review of Sociology* 48 (1): 65–85. <https://doi.org/10.1146/annurev-soc-090221-035954>.

Murphy, Mary C., Amanda F. Mejia, Jorge Mejia, et al. 2020. "Open Science, Communal Culture, and Women's Participation in the Movement to Improve Science." *Proceedings of the National Academy of Sciences* 117 (39): 24154–64. <https://doi.org/10.1073/pnas.1921320117>.

National Academies of Sciences, Engineering, and Medicine (NASEM). 2018. *Open Science by Design: Realizing a Vision for 21st Century Research*. Washington, DC: National Academies Press. <https://doi.org/10.17226/25116>.

National Center for Science and Engineering Statistics (NCSES). 2023. *Doctorate Recipients from U.S. Universities: 2022*. NSF 24-300. Alexandria, VA: National Science Foundation. <https://nces.nsf.gov/pubs/nsf24300>.

Nosek, Brian. 2019. "Strategy for Culture Change." <https://www.cos.io/blog/strategy-for-culture-change>.

Nosek, Brian A., Tom E. Hardwicke, Hannah Moshontz, et al. 2022. "Replicability, Robustness, and Reproducibility in Psychological Science." *Annual Review of Psychology* 73 (1): 719–48. <https://doi.org/10.1146/annurev-psych-020821-114157>.

Piwowar, Heather, Jason Priem, Vincent Larivière, et al. 2018. "The State of OA: A Large-Scale Analysis of the Prevalence and Impact of Open Access Articles." *PeerJ* 6 (February): e4375. <https://doi.org/10.7717/peerj.4375>.

R Core Team. 2026. *R: A Language and Environment for Statistical Computing. Version 4.5.3*. Vienna: R Foundation for Statistical Computing. <https://doi.org/10.32614/R.manuals>.

Robson, Samuel G., Myriam A. Baum, Jennifer L. Beaudry, et al. 2021. "Promoting Open Science: A Holistic Approach to Changing Behaviour." *Collabra: Psychology* 7 (1): 30137. <https://doi.org/10.1525/collabra.30137>.

Ross-Hellauer, Tony, Stefan Reichmann, Nicki Lisa Cole, Angela Fessler, Thomas Klebel, and Nancy Pontika. 2022. "Dynamics of Cumulative Advantage and Threats to Equity in Open Science: A Scoping Review." *Royal Society Open Science* 9 (1): 211032. <https://doi.org/10.1098/rsos.211032>.

Settles, Isis H., Leah R. Warner, NiCole T. Buchanan, and M. K. Jones. 2020. "Understanding Psychology's Resistance to Intersectionality Theory Using a Framework of Epistemic Exclusion and Invisibility." *Journal of Social Issues* 76 (4): 796–813. <https://doi.org/10.1111/josi.12403>. NiCole T. Buchanan, and Kristie Dotson, "Epistemic Exclusion: Scholar(ly) Devaluation That Marginalizes Faculty of Color," *Journal of*

Diversity in Higher Education — please verify author list, title and year before submission.]

Smart, Pippa. 2022. "The Evolution, Benefits, and Challenges of Preprints and Their Interaction with Journals." *Science Editing* 9 (1): 79–84.  
<https://doi.org/10.6087/kcse.269>.

Steel, Brent, Peter List, Denise Lach, and Bruce Shindler. 2004. "The Role of Scientists in the Environmental Policy Process: A Case Study from the American West." *Environmental Science & Policy* 7 (1): 1–13.  
<https://doi.org/10.1016/j.envsci.2003.10.004>.

US News and World Report. 2022. "Best Colleges 2022: About the Rankings/Methodology." US News and World Report.  
<https://www.usnews.com/education/best-colleges/articles/rankings-methodologies>.

Weeden, Kim A. 2023. "Crisis? What Crisis? Sociology's Slow Progress Toward Scientific Transparency." *Harvard Data Science Review* 5 (4).  
<https://doi.org/10.1162/99608f92.151c41e3>.

Wickham, Hadley. 2025. *Rvest: Easily Harvest (Scrape) Web Pages*. R package version 1.0.5. <https://rvest.tidyverse.org/>.

## Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [SupplementarymaterialsRhetoricsNHB.docx](#)